

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1793

Willow Creek 138 kV
February 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1793. The Interconnection Customer proposes a 123.8 MW solar photovoltaic facility interconnecting at the Willow Creek 138 kV substation in the Virginia area. The request was received on June 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 513 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,852,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Golden Hills to Willow Crk 138kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 163.5% (Rating: 240 MVA)
DFAX: 19.9%
Pre-project loading: 153.2%
2. Facility: Golden Hills to Instituto Tecnologico De Ponce 138kV
Contingency: Louisa Generation Facility Substation to Fluvanna P.S 345kV L
Loading: 161.4% (Rating: 238 MVA)
DFAX: 19.9%
Pre-project loading: 151.1%
3. Facility: Abbeville to Instituto Tecnologico De Ponce 138kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 159.7% (Rating: 236 MVA)
DFAX: 19.9%
Pre-project loading: 149.3%
4. Facility: Apple Data Ctr Pv to Fern Vly 138kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 158.6% (Rating: 223 MVA)
DFAX: 19.9%
Pre-project loading: 147.5%
5. Facility: Fern Vly to Cabrillo 138kV
Contingency: Louisa Generation Facility to Fluvanna P.S 345kV
Loading: 158.5% (Rating: 227 MVA)
DFAX: 19.9%
Pre-project loading: 147.7%

6. Facility: Abbeville to Cabrillo 138kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 157.8% (Rating: 235 MVA)
DFAX: 19.9%
Pre-project loading: 147.3%
7. Facility: Cold Spring Rd to Apple Data Ctr Pv 138kV
Contingency: Fluvanna P.S Substation to Louisa Generation Facility 345kV
Loading: 157.2% (Rating: 210 MVA)
DFAX: 19.9%
Pre-project loading: 145.5%
8. Facility: Cold Spring Rd to N Anna 138kV Line
Contingency: Fluvanna P.S to Louisa Generation Facility Substation 345kV
Loading: 156.6% (Rating: 211 MVA)
DFAX: 19.9%
Pre-project loading: 144.9%
9. Facility: Franklin-Laguna to Ellinwood 138kV Line
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 128.7% (Rating: 262 MVA)
DFAX: 8.9%
Pre-project loading: 124.5%
10. Facility: Orrington Substation to Unisource Golden Valley 138kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV Line
Loading: 125.6% (Rating: 250 MVA)
DFAX: 8.9%
Pre-project loading: 121.1%
11. Facility: Orrington to Ellinwood 138kV line
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV L
Loading: 116.0% (Rating: 285 MVA)
DFAX: 8.9%
Pre-project loading: 112.1%
12. Facility: Bear Island to Dunn Siding 345kV
Contingency: Leavitt to Birchwood 345kV
Loading: 112.7% (Rating: 2,769 MVA)
DFAX: 6.8%
Pre-project loading: 112.4%
13. Facility: Littlefork to Lake Lure 138kV Line
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 105.4% (Rating: 918 MVA)
DFAX: 15.0%
Pre-project loading: 103.3%
14. Facility: Conservation to Bremon Bluff 138kV line
Contingency: Fluvanna P.S to Louisa Generation Facility 345kV
Loading: 105.0% (Rating: 244 MVA)
DFAX: 14.0%
Pre-project loading: 97.9%
15. Facility: Thomaston to Carrabelle 345kV
Contingency: Thomaston to Carrabelle 345kV Ckt 2
Loading: 102.4% (Rating: 1,435 MVA)
DFAX: 8.0%
Pre-project loading: 101.7%
16. Facility: Thomaston to Carrabelle 345kV
Contingency: Thomaston to Carrabelle 345kV Ckt 1

Loading: 102.4% (Rating: 1,435 MVA)

DFAX: 8.0%

Pre-project loading: 101.7%

17. Facility: Louisa Generation Facility to Gordonsville Energy Lp 138kV #1

Contingency: Fluvanna P.S to Louisa Generation Facility 345kV

Loading: 102.0% (Rating: 378 MVA)

DFAX: 5.6%

Pre-project loading: 100.2%

18. Facility: Louisa Generation Facility to Gordonsville Energy Lp 138kV Cir. 2

Contingency: Fluvanna P.S to Louisa Generation Facility 345kV

Loading: 102.0% (Rating: 378 MVA)

DFAX: 5.6%

Pre-project loading: 100.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Golden Hills to Willow Crk 138kV	Rebuild line with higher-capacity conductor	\$4,419,000	100% (\$4,419,000)
Golden Hills to Instituto Tecnologico De Ponce 138kV	Rebuild line with higher-capacity conductor	\$5,478,000	100% (\$5,478,000)
Abbeville to Instituto Tecnologico De Ponce 138kV	Rebuild line with higher-capacity conductor	\$10,019,000	100% (\$10,019,000)
Apple Data Ctr Pv to Fern Vly 138kV	Rebuild line with higher-capacity conductor	\$9,899,000	100% (\$9,899,000)
Cabrillo to Fern Vly 138kV	Rebuild line with higher-capacity conductor	\$11,389,000	100% (\$11,389,000)
Abbeville to Cabrillo 138kV	Rebuild line with higher-capacity conductor	\$3,713,000	100% (\$3,713,000)
Cold Spring Rd to Apple Data Center Pv 138kV	Rebuild line with higher-capacity conductor	\$3,118,000	100% (\$3,118,000)
Cold Spring Rd Substation to N Anna 138kV	Rebuild line with higher-capacity conductor	\$3,224,000	100% (\$3,224,000)
Franklin-Laguna to Ellinwood 138kV Line	Rebuild line with higher-capacity conductor	\$28,211,000	100% (\$28,211,000)
Orrington to Unisource Golden Vly 138kV Line	Rebuild line with higher-capacity conductor	\$4,705,000	100% (\$4,705,000)
Ellinwood Substation to Orrington 138kV	Rebuild line with higher-capacity conductor	\$5,975,000	100% (\$5,975,000)
Bear Island to Dunn Siding 345kV	Rebuild line with higher-capacity conductor	\$54,892,000	0% (\$129,000)
Littlefork to Lake Lure 138kV Line	Reconductor line	\$1,827,000	4% (\$70,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Conservation to Bremo Bluff 138kV line	Reconductor line	\$10,489,000	100% (\$10,489,000)
Thomaston to Carrabelle 345kV	Reconductor line	\$21,635,000	2% (\$374,000)
Thomaston to Carrabelle 345kV	Reconductor line	\$27,509,000	2% (\$475,000)
Gordonsville Energy Lp to Louisa Generation Facility 138kV Cir. 1	Reconductor line	\$716,000	10% (\$74,000)
Louisa Generation Facility to Gordonsville Energy Lp 138kV Cir. 2	Reconductor line	\$884,000	10% (\$91,000)
	Total allocated to project		\$101,852,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.